The central kernel in a symmetric action on pairs

Human Compression project: a component of the L044 recovery

The acquired faithful-lift reduction leaves the following exact case. The natural alternating group A_{2m} has a string C-marking $\rho_0, \dots, \rho_{m-1}$, with $m \geq 8$ even. Its omission $H = \langle \rho_1, \dots, \rho_{m-1} \rangle$ is transitive on a system of m pairs and induces S_m. Put

$$B = \langle \rho_1, \dots, \rho_{m-2} \rangle, \quad V = \mathbb{F}_2^m, \quad z = (1, \dots, 1), \quad E = \{v \in V : \sum v_i = 0\}.$$

The acquired reduction gives

$$B \cap V = \langle z \rangle, \qquad H \cap V = E. \tag{1}$$

The images $s_1, \dots, s_{m-2}$ of the marks of B generate S_m independently. They form a string of involutions, but no intersection property is assumed for them. The extra mark ρ_{m-1} is independent of B. We acquire the full actual pair-centralizer lemma, the corrected primitive independent-string bound, and B5's full irredundant bound: every transitive imprimitive group of degree $q \geq 9$ has maximum irredundant length at most $q - 4$. These bounds apply before any quotient IP has been established.

Lemma 1. *Let $s_1, \dots, s_{q-2}$ be independent involutions generating S_q, where $q \geq 6$. Suppose they form a string and every maximal omission is intransitive in the natural action. Then, up to simultaneous conjugacy and reversal, they are*

$$(2\ 3), \quad (1\ 2)(3\ 4), \quad (4\ 5), \quad (5\ 6), \dots, (q-1\ q). \tag{2}$$

No intersection property is needed.

Proof. The noncommutation graph is connected. Otherwise its components generate commuting nontrivial normal subgroups of S_q. Their intersection is central, hence trivial, giving a nontrivial direct decomposition. This is impossible: any nontrivial normal subgroup contains A_q, whose centralizer in S_q is trivial. Since all far-apart marks commute, the noncommutation graph is therefore the path on the given order of marks.

For each label choose an edge joining distinct orbits of its omission. The acquired two-component fracture-forest lemma applies literally: its proof uses only these orbit separations and the matching condition on each label, not IP. In detail, the selected $q - 2$ edges form a forest, since a cycle would bypass a selected edge with other labels. An extra edge joins the two components; otherwise its selected-tree path bypasses some differently labelled selected edge. Two extra edges must intersect. If they do not have the same endpoints, the two selected paths force one alternating square, whose labels each occur twice and which admits no other extra edge. Thus the full graph is either a tree with one repeated label, a tree with several repeated labels sharing one edge, or one alternating square with uniquely labelled attached trees. We now impose the string condition on these exact possibilities.

In the tree case with several repeated labels, write the common edge as $\tau = \{a, b\}$. Every repeated mark is $\tau\sigma_i$, where σ_i is its other edge, disjoint from τ. Every edge incident to τ is a single transposition and fails to commute with every repeated mark. Consequently there are at most two repeated labels. If there are two, call them i, j. Every connector incident to τ is a common neighbour of i, j in the noncommutation path. A path has at most one such neighbour,

so there is exactly one connector k. Moreover i, j commute, since otherwise i, k, j would be a triangle; hence σ_i, σ_j are disjoint. Removing k leaves τ on its own two vertices, and both other edges on the opposite side. The unique tree path from σ_i to σ_j avoids k and τ. Its intervening edges are single transpositions. Its consecutive labels therefore give a noncommutation path from i to j avoiding k. Together with i, k, j this is a cycle, impossible. Multiple repeated labels are excluded.

There is now one repeated mark, with two disjoint edges. If there were at least two intervening tree edges between them, those single transpositions and the repeated mark would again form a cycle in the noncommutation graph. Thus exactly one single connector lies between them. They form a path on four vertices. An edge leaving a middle vertex of this path fails to commute with both core marks, giving a triangle. An edge leaving an outer vertex fails to commute with the double transposition. That mark already has the connector as a neighbour, so at most one edge leaves the core. Connectedness and $q > 4$ supply exactly one. The attached tree has no vertex of degree three, since its three incident transpositions would give a triangle. Thus it is a path attached at an outer vertex. This gives (2), and the noncommutation path has only its two opposite string orders.

Finally, in the alternating-square case the two core marks commute. Every edge leaving the square is a single transposition noncommuting with both. There can be only one such edge k, as two would give a cycle in the noncommutation graph. The vertex k already has its two core neighbours, so no further edge can meet its outside endpoint. The connected full graph then has only five vertices, contrary to $q \geq 6$. □

Lemma 2. *In the standing central-kernel case, if $m \geq 10$, every maximal omission of $s_1, \ldots, s_{m-2}$ is intransitive.*

Proof. Such an omission is a proper subgroup of S_m, with an independent generating string of length $m - 3$. If it were transitive imprimitive, the acquired irredundant bound would give length at most $m - 4$. If it were primitive other than A_m, S_m, the corrected independent-string bound would give the same contradiction. The only possible transitive omission is therefore A_m.

An interior omission is generated by two commuting nontrivial subgroups, on the nonempty intervals on its two sides. They are normal in the group they generate. They cannot generate the simple group A_m, since simplicity would make both that whole group, which is nonabelian.

At the first endpoint put $T = \langle \rho_2, \ldots, \rho_{m-2} \rangle$. The omitted ρ_0 centralizes T. If its block image were A_m, the acquired actual pair-centralizer lemma would force ρ_0 to preserve the pairs: this primitive block group is not its regular prime-dihedral exception. The whole parent would then preserve the pairs, a contradiction. At the other endpoint put $F = \langle \rho_1, \ldots, \rho_{m-3} \rangle$. The extra mark ρ_{m-1} centralizes F. An A_m block image has trivial centralizer in S_m, so that mark would belong to V. A flip vector commuting with the transitive A_m is constant, hence is either 1 or z. Both belong to B, contrary to its independence. This excludes the last possible transitive omission. □

Lemma 3. *An independent connected string of six involutions generating literal S_8 is conjugate to (2) or its reverse.*

Proof. This is the bounded exact premise supplied with this manuscript. Its engine is the acquired corrected primitive independent-string engine, with the ambient group changed to literal S_8 and its single target and maximum length both six. It lists every nonidentity involution (there are 763) and extends prefixes by involutions commuting with all but their last mark, and noncommuting with that last mark. It rejects a prefix only if one member is already redundant, it has already generated S_8 too soon, it has reached length six, or its subgroup index is less than 2^{6-j} at length j. Every rejection is necessary for an independent completion of length six. It takes conjugation orbits under the stabilizer of the entire fixed prefix, beginning with S_8 itself.

These operations preserve every completion condition and therefore preserve exhaustiveness. In particular there is no IP test, no end-intersection test, and no centralizer-quotient pruning.

The complete visited-prefix counts are

$$1, 4, 63, 1308, 934, 113, 3.$$

There are exactly two ordered classes at length six, with types $[4, 6, 3, 3, 3]$ and $[3, 3, 3, 6, 4]$. A separate exact check verifies generation of literal S_8, independence under all six omissions, all involution and string conditions, and simultaneous conjugacy of each tuple to (2) or its reverse. It also checks that the two ordered classes differ and rejects a redundant repeated-generator control. The trusted premises are the acquired enumeration argument and GAP's exact finite group operations. The checker pins its input bytes, rejects optimized Python, clears that variable for children and requires completed executions with no GAP errors. This is a complete length-six independent-string boundary, not an inference from a C-group census. □

Theorem 4. *The standing central-kernel case is impossible, for every even $m \geq 8$.*

Proof. The independent projected string has connected noncommutation graph by the direct-decomposition argument in Lemma 1. For $m \geq 10$, Lemmas 2 and 1 put it in form (2); at $m = 8$ use Lemma 3. We retain either orientation relative to the omitted mark ρ_0.

First, ρ_0 does not commute with z. Otherwise it and H would centralize z, so all of A_{2m} would do so, whereas its natural centralizer is trivial. Since ρ_0 centralizes $T = \langle \rho_2, \ldots, \rho_{m-2} \rangle$, we have $z \notin T$. By (1), projection is consequently an isomorphism from T to its block image. This is actual faithfulness; no IP is asserted for the whole projected marking.

If ρ_0 is next to the tail end of (2), its commuting subgroup T fixes one original pair setwise and induces S_{m-1} on the others. The acquired one-fixed-pair lemma forces ρ_0 to preserve every pair, a contradiction. We may thus place it next to the core end. Its commuting subgroup T now projects isomorphically to

$$S_{\{1,2\}} \times S_{\{3,\ldots,m\}}. \tag{3}$$

Indeed, if $u = (1\ 2)(3\ 4)$ and $v = (4\ 5)$, then $(uv)^3 = (1\ 2)$; this recovers $(3\ 4)$ and all the remaining path transpositions. Write $b = m - 2$, so $b \geq 6$ is even.

The preimage in T of the pure A_b factor in (3) fixes the four points over $\{1, 2\}$: its action there takes values in the abelian flip group C_2^2, whereas A_b is perfect. It has no fixed point over the other b pairs, since it is transitive on those pairs. Its fixed set is therefore exactly these four points. Thus ρ_0 preserves the four points and their complement separately.

On the $2b$ points the actual induced group of T has primitive block image S_b. The pair-centralizer lemma forces the restriction of ρ_0 to preserve those pairs. Its induced block permutation centralizes S_b, so is trivial; its flip vector is constant. The restriction is consequently either 1 or the flip z_b of all b pairs. Both permutations are even because b is even. The restriction of ρ_0 to the four other points must therefore also be even.

The pure S_2 element in (3) has an involutory preimage in T, by faithfulness. On those four points it exchanges the two original pairs, and hence is a double transposition. The even centralizer of a double transposition in S_4 is the Klein four group of double transpositions. It preserves every partition of the four points into pairs. Since ρ_0 centralizes this element and is even on these points, it preserves their two original pairs as well. Thus all of ρ_0 preserves the pair system, contradicting $\langle H, \rho_0 \rangle = A_{2m}$. This proves the theorem. □

Attribution and scope. The source supplies the invariant-module reduction in its Section 3.2. The present argument uses the already acquired full irredundant bound for imprimitive groups and the corrected primitive bound to justify every projected omission. It derives the small core directly from the string condition and the acquired fracture forest, without quotient IP; the actual centralizer and parity argument then closes the surviving kernel case. The degree-eight

independent-string census is new and distinct from the earlier C-group census. No splitting of B as $C_2 \times S_m$ is assumed. Together with the accepted faithful component this recovers the full symmetric block-image branch. General imprimitive cases, the final strong-induction assembly and whole-source review remain separate obligations. No whole-original, conjecture, Lean or compression credit is granted by this component.